

Databases are essential for storing structured information.
Each table has a unique name and a defined set of columns.
Indexes accelerate data retrieval by avoiding full table scans.
Queries allow selective reading of rows matching user criteria.

```
def fetch_user(id):  
    user = db.query("SELECT * FROM users WHERE id = ?", id)  
    if user is None:  
        raise NotFound("user not found")  
    return user
```

Indexes improve query speed

```
class UserService:  
    def __init__(self, db):  
        self.db = db  
  
    def list(self):  
        return db.query("SELECT id, name FROM users")
```

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Stored procedures encapsulate business logic on the server.
Triggers execute automatically in response to data changes.
Views provide named queries that simplify complex joins.
Roles and grants control access to sensitive data.

```
def get_user(id):  
    async with pool.acquire() as conn:  
        row = await conn.fetchrow(  
            "SELECT id, name FROM users WHERE id = $1", id  
        )  
        if not row:  
            raise HTTPError(404)  
        return dict(row)
```

Stored procedures and triggers

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